

UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 1: Signals: Analog, Discrete-Time, and Digital

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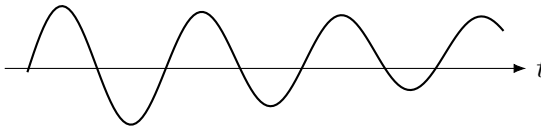
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1 What is a Digital Signal? (1D Case)

We distinguish three flavors of one-dimensional signals, based on how the independent variable (time) and the dependent variable (amplitude) are represented.

1.1 Continuous-Time (“Analog”) Signals

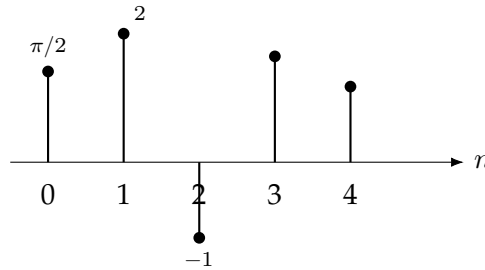
A continuous-time signal $x_a(t)$ is a real- or complex-valued function defined for all real-valued t .



Examples: speech signals, EEG, images formed with focused light.

1.2 Discrete-Time Signals

A discrete-time signal $x[n]$ is defined only for integer-valued n .



Discrete-time signals arise in (at least) two ways:

- **Sampled continuous signals.** $x[n] = x_a(nT)$, e.g. a sampled audio signal.
- **Inherently discrete signals.** Quantities that only update at discrete instants, e.g. the daily maximum temperature, or the number of students late to class on days when class meets.

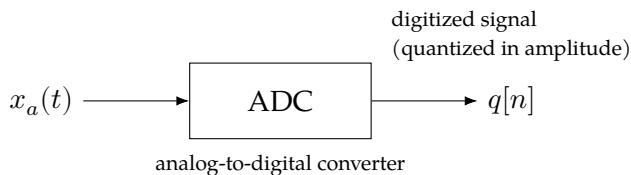
1.3 Digital Signals

Definition (Digital Signal).

A **digital signal** is sampled in time *and* in amplitude: each value $x[n]$ is stored with a finite number of bits, so the numbers are rounded. This is called **finite precision** or **quantization**.

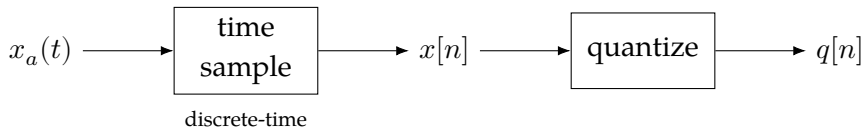
2 From the Real World to Numbers

2.1 The Real ADC



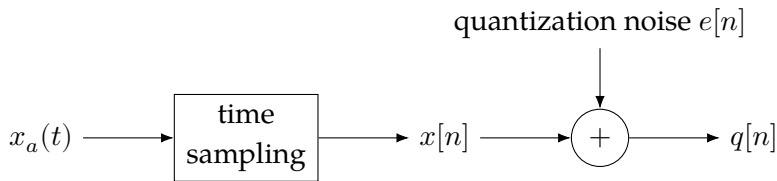
2.2 A Useful Model

We conceptually split the ADC into a time-sampling stage followed by an amplitude-quantization stage:



2.3 An Even More Useful Model

Quantization is modeled as *additive quantization noise* $e[n]$ injected after ideal time-sampling:



Remark.

Discrete-time systems *can* be linear. **Digital systems cannot be linear**; quantization is an inherently nonlinear operation.

3 DSP Topics for the Course

- **Sampling and reconstruction** (ADC and DAC)
 - Nyquist rate
- **Discrete-time linear systems**

$$x_a(t) \rightarrow \boxed{\text{linear system}} \rightarrow y_a(t), \quad x[n] \rightarrow \boxed{\text{linear system}} \rightarrow y[n]$$

- **Signal representation and coding**
 - Discrete Time Fourier Transform (DTFT)
 - Discrete Fourier Transform (DFT)
 - Discrete cosine transform and wavelets (WebP, JPEG, MPEG, MP3, etc.)
- **Implementation**
 - Design of digital filters

- Hardware (CPU, DSP, GPU, FPGA)
- Computational cost (algorithmic) and quantization / overflow (configuration details)

4 Why DSP Instead of Analog Signal Processing?

- Cost
- Storage of data
- Availability
- Easily reconfigured
- Implementation time
- Reliability
- Flexibility

Optional Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 1, 2.1, 2.2, 2.4
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 1.1–1.3